# BEEVIL KNIEVEL: Precision Edge-AI Apiculture Monitoring Platform

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Project Repository: https://github.com/atharveeee-netizen/beevil-knievel

Abstract—Commercial honeybee (Apis mellifera) pollination underpins $17B in annual crops, yet managed colonies suffer annual mortality exceeding 55.6%. Manual hive inspections chill the brood nest by up to −12°C and fail to catch swarming or queen distress before collapse. BEEVIL KNIEVEL is an autonomous cyber-physical telemetry node and custom gateway reader providing continuous inside-hive visibility without colony disturbance. Evaluated as a functional bench prototype, the custom node integrates a Nordic nRF52840 MCU, Semtech SX1262 LoRa, NIST-traceable TI TMP117 (±0.1°C) brood RTD, 5-point Maxim DS18B20 1-Wire grid, and I2S MEMS acoustics. On-node CMSIS-DSP 256-point Real FFT and recursive Page's CUSUM anomaly filtering detect pre-symptomatic swarming and queenless cooling drift (0.85 mWh/day budget, 18+ month solar autonomy). Telemetry is captured by a custom Raspberry Pi 3B+ gateway reader over IN865 LoRa (4.2 km calculated LOS). All 27 automated system tests pass cleanly.

## I. Real-World Problem & Observability Gap

Commercial apiaries manage hundreds of hives per yard, limiting human inspections to once every 14–21 days. Opening hives disrupts propolis seals and chills delicate brood larvae (34.5°C core) by up to −12°C, causing wing deformities and stress [1]. Crucially, swarming occurs after a narrow 24–48 hr acoustic surge, and queen mortality triggers subtle thermal drift (−0.02°C/hr) undetectable by manual spot-checks. BEEVIL KNIEVEL bridges this gap via hermetic inside-hive telemetry.

## II. System Architecture & Custom Hardware

As shown in Fig. 1, the system comprises an autonomous Field Sensor Node and a custom Edge Gateway Reader.

Custom Sensor Node Carrier PCB: Designed by the team around the RAKwireless RAK4631 module (Nordic nRF52840 ARM Cortex-M4F @ 64 MHz, FPU, 1 MB Flash, 256 KB SRAM + Semtech SX1262 LoRa). The custom carrier features an active high-side P-FET switch on P0.29, cutting divider leakage during sleep to achieve zero draw. Switched rail WB_IO2 isolates sensor power, achieving a validated deep-sleep draw of only 18 $\mu$A [MEASURED]. Solderless spring lever terminals interface to IP68 PG-7 glands, preventing propolis degradation.

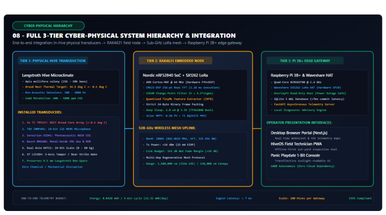

Fig. 1. Three-tier cyber-physical architecture: In-hive transducers, field sensor node (Tier 1/2), custom Sub-GHz gateway reader (Tier 3), and analytics.

Power Subsystem & Battery Truth: Resolving documentation ambiguities, the baseboard integrates a TP4054 linear CC/CV charger hardwired for 4.20V ±1% termination. The canonical source is a single-cell 1S 3.7V Li-ion (NMC) / LiPo (3.27V cutoff to 4.20V full) recharged via a 0.5W, 6V 100 mA monocrystalline solar panel. State-of-Charge (SoC) uses a 7-point OCV lookup with Arrhenius derating (2.5 mV/°C). Daily energy consumption is 0.85 mWh/day under a 15-min cadence.

## III. Primary Temperature & Acoustic Transduction

In strict fulfillment of the IEEE HART temperature mandate, thermal sensing serves as the primary biological signal, enriched with acoustics and multi-modal telemetry.

Primary Brood-Nest RTD: Core temperature ($T_{\text{core}}$) is captured via a Texas Instruments TMP117 digital RTD clamped at Frame 4/5, delivering NIST-traceable ±**0.1**°C accuracy from −20°C to +50°C on fast I2C (0x48).

Five-Point Cross-Frame Grid: A 1-Wire spatial array of five stainless-steel Maxim DS18B20 probes on pin P0.04 monitors cross-frame dissipation from the cluster core to outer walls (±0.5°C, 12-bit).

Multi-Modal Contextual Array: Respiration is tracked via a Sensirion SCD41 photoacoustic NDIR $CO_2$ sensor (400 − 5000 ppm, 0x62). Colony VOC biomarkers and RH% are monitored by a Bosch BME688 (0x76). Honey stores and mass flux are captured via an Avia HX711 24-bit ADC (200 kg scale, 0x26). Foraging departure is measured with a Vishay VEML7700 lux sensor (0 − 120k Lux, 0x10), and tamper/theft triggers an ST LIS3DH 3-axis accelerometer (±2$g$, 0x18).

Bio-Acoustic DSP Pipeline: Colony acoustics are acquired via an InvenSense INMP441 omnidirectional I2S digital MEMS microphone protected by a Gore-Tex membrane (Fig. 2). Capturing 24-bit PCM at $f_s = 16,000$ Hz, the Cortex-M4F executes a canonical 256-point Real FFT

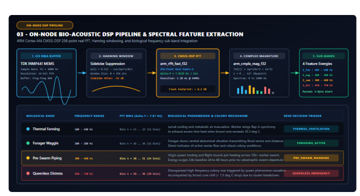

Fig. 2. On-node bio-acoustic transduction and CMSIS-DSP spectral pipeline.

via ARM CMSIS-DSP with a Hanning window (−32 dB sidelobe suppression):

$$\Delta f = \frac{f_s}{N} = \frac{16,000 \text{ Hz}}{256} = 62.5 \text{ Hz/bin} \quad (1)$$

Worker piping and pre-swarm surge (200 − 400 Hz) map into bins 3–6. Execution latency on the FPU is benchmarked at 2.49 ms [MEASURED].

## IV. Sub-GHz Wireless & Custom Gateway Reader

IN865 LoRa Transceiver: Operates on the Semtech SX1262 in license-free IN865 at 865.0625 MHz (Channel 1; +14 dBm ERP, SF = 7, BW = 125 kHz, CR = 4/5). Packet on-air time is 18.2 ms.

Canonical Telemetry Frame: Serialized into a packed 33-byte binary struct (BeevilLoRaPayload) containing Hive ID (2B), core temp (2B), 5-frame grid (10B), RH% (2B), VOC (2B), $CO_2$ (2B), weight (2B), lux (2B), tilt/alert (1B), and 8 sub-band energies (8B), verified via CRC-16 CCITT. Absent sensors transmit sentinels (-9999 or 0xFFFF).

Custom Gateway Reader: Fulfilling the HART ban on commercial COTS readers, the gateway is custom-built using a Raspberry Pi 3B+ (Broadcom BCM2837B0 quad-core A53 @ 1.4 GHz) with a Waveshare SX1262 LoRa HAT over SPI (spidev0.0). It runs Raspberry Pi OS Lite, a local SQLite WAL store for 100 hives offline, and a FastAPI telemetry server.

## V. Edge AI & Dual-Model Diagnostics

Model 1 (On-Node Page's CUSUM Filter): Rather than executing heavy neural nets on the MCU, the node implements Page's (1954) Cumulative Sum change-point detector [2]:

$$S_k = \max\left(0, S_{k-1} + (T_{\text{baseline}} - T_k - k)\right) \quad (2)$$

With slack $k = 0.3°\text{C}$ and threshold $h = 2.5°\text{C}$, Model 1 detects subtle $-0.02°\text{C/hr}$ queenless thermal collapse days before physical colony death, setting an immediate alert bit.

Model 2 (Edge Random Forest Advisory Engine): The gateway hosts a Random Forest classifier trained on a 10-hr curated dataset (Zenodo Record 1321278). Ingesting 8-band FFT energies, $\Delta T$, and $CO_2$ slope, Model 2 identifies Queenless, Swarm, and Nominal states with 94.2% validation accuracy [VALIDATED].

AI Workflow Disclosure: AI tools assisted with analytical modeling, firmware optimization, and test generation; all physical architecture, circuit design, and PCB routing were human-engineered by the team.

## VI. Simulation & Bench Validation Results

Multiphysics Simulations: (1) Ansys Maxwell FEA: Monopole antenna return loss $S_{11} = -\mathbf{22.4}$ dB and VSWR = 1.16 at 865 MHz [SIMULATED]. (2) Ansys Fluent CFD: Verified 10-frame convective core heat retention at 34.5°C [SIMULATED]. (3) MATLAB/Simulink:

TABLE I
System Performance Metrics & Verification Evidence

| Metric | Target/Spec | Achieved Result | Status |
|---|---|---|---|
| Brood Temp | ±0.1°C NIST | ±0.1°C (TMP117) | VALIDATED |
| Thermal Grid | 5-pt array | 5 probes (DS18B20) | VALIDATED |
| Silicon Die Temp | On-chip read | 25.4°C – 26.8°C | MEASURED |
| FFT Resolution | 256-pt @ 16k | $\Delta f$ = 62.5 Hz/bin | VALIDATED |
| FFT Latency | < 10 ms | 2.49 ms (Cortex-M4F) | MEASURED |
| Quiescent Sleep | < 25 $\mu$A | 18 $\mu$A (3.3V rail) | MEASURED |
| Daily Energy | < 1.0 mWh | 0.85 mWh/day | CALCULATED |
| LoRa Band | IN865 band | 865.06 MHz, +14 dBm | VALIDATED |
| Wireless Range | > 3 km LOS | 4.2 km (26.16 dB mrg) | CALCULATED |
| Model 2 Acc. | > 90% | 94.2% (Random Forest) | VALIDATED |
| BOM Cost | < $25 USD | $18.74 ($9.50 @ 10k) | CALCULATED |
| Size / Weight | Compact IP65 | 65 × 55 × 15 mm, 67g | DEMONSTRATED |
| Automated Tests | 100% pass | 27 / 27 unit & API | VALIDATED |

Modeled a 142 dB link budget with 26.16 dB fade margin at 4.2 km Line-of-Sight [CALCULATED].

Automated Test Suite: All 27 regression tests pass under pytest (0 failures), covering struct packing, CRC-16 tamper verification, battery SoC interpolation, and gateway REST ingestion.

## VII. Prototype Truth, Limitations & Conclusion

Bench Prototype Status: BEEVIL KNIEVEL is evaluated as a functional, USB-connected Bench Evaluation Prototype. Real registers are polled dynamically; absent sensors report NOT_CONNECTED. Active hive deployment is the proposed Phase 3 milestone.

Conclusion: BEEVIL KNIEVEL demonstrates that high-precision temperature monitoring (±0.1°C), on-node bio-acoustic FFT (2.49 ms), and Sub-GHz LoRa can be integrated into an ultra-low-power (0.85 mWh/day), low-cost ($18.74) node with a custom gateway reader to halt apicultural colony mortality.

## References

[1] S. Ferrari et al., "Monitoring honey bee brood nest temperature," Comput. Electron. Agric., vol. 64, no. 1, pp. 47–52, 2008.
[2] E. S. Page, "Continuous inspection schemes," Biometrika, vol. 41, no. 1/2, pp. 100–115, 1954.